

R 4.2.12 Considere la función $f: \mathbb{R} - \{-5\} \rightarrow \mathbb{R}$, tal que $f(x) = \frac{x}{(x+5)^2} + \frac{6}{(x+5)^3}$. Determine el mayor subconjunto del dominio de f en el cual $f(x) \leq 0$.

Paso 1: Dominio $D_f = \mathbb{R} - \{-5\}$
 $x+5 \neq 0$
 $x \neq -5$

Paso 2: Unificar en 1 fraccion

$$f(x) = \frac{x}{(x+5)^2} + \frac{6}{(x+5)^3} \quad \frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$$

$$\frac{(x+5)}{(x+5)} \cdot \frac{x}{(x+5)^2} + \frac{6}{(x+5)^3} \quad \frac{a}{b} + c \cdot \frac{b}{b}$$

$$\frac{x(x+5)}{(x+5)^3} + \frac{6}{(x+5)^3} \quad \frac{d}{d} \frac{a}{b} + \frac{c}{d} \cdot \frac{b}{b}$$

$$\frac{x^2 + 5x + 6}{(x+5)^3}$$

$$\frac{da}{db} + \frac{cb}{db} = \frac{da+cb}{db}$$

$$\begin{array}{r} x^2 + 5x + 6 \\ x \quad \quad \quad 3 = 3x \\ x \quad \quad \quad 2 = 2x \\ \hline \quad \quad \quad 5x \end{array}$$

$$x^a \cdot x^b = x^{a+b}$$

$$x^2 \cdot x^3 = x^{2+3} = x^5$$

$$x^2 + 5x + 6 = (x+3)(x+2)$$

$$x^2 + 5x + 6 = (x+3)(x+2)$$

$$\frac{x^2 + 5x + 6}{(x+5)^3}$$

$$\frac{(x+3)(x+2)}{(x+5)^3} \leq 0$$

Paso 3: Poner puntos críticos

$$\begin{array}{lll} x+3=0 & x+2=0 & x+5 \neq 0 \\ x=-3 & x=-2 & x \neq -5 \end{array}$$

	$-\infty$	-5	-3	-2	$+\infty$
$x+5$	—	●	+	+	+
$x+3$	—	—	●	+	+
$x+2$	—	—	—	●	+
$f(x)$	—	+	—	+	+

$$f(x) \leq 0?$$

$$f(x) \leq 0 :]-\infty, -5[\cup [-3, -2]$$

$$1) f(x) = 7 \sqrt{\frac{x-3}{x^2-25}}$$

$$7 \sqrt{\frac{x-3}{x^2-25}} \geq 0$$

$$\frac{x-3}{x^2-25} \geq 0$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$\frac{x-3}{(x-5)(x+5)} \geq 0 \quad \begin{array}{l} x^2-25 \\ = x^2-5^2 \\ = (x-5)(x+5) \end{array}$$

$$\begin{array}{lll} x-3=0 & x-5 \neq 0 & x+5 \neq 0 \\ x=3 & x=5 & x=-5 \end{array}$$

	$-\infty$	-5	3	5	$+\infty$
$x+5$	—	• +	—	• +	—
$x-3$	—	—	• +	—	• +
$x-5$	—	—	—	• +	—
*	—	+	—	+	—

$$f(x) \geq 0$$

$$D_f:]-5, 3] \cup]5, +\infty[$$

Se define la función $f(x) = |2x - 9| - 2$

Donde es positiva?

$$|2x-9|-2 \geq 0$$

$$2x - 4 \geq 0$$

$$2x \geq 4$$

$$x \geq 2$$

$$[2, +\infty[$$

$$|2x - 7|$$

$$2x - 7 \text{ si } x \geq 0$$

$$-(2x-9) \text{ si } x < 0$$

$$-2x + 9$$

$$2x - 420$$

$$2x \subset \mathcal{C}$$

x 22

$$3-\infty, 2[$$

$$-\infty \quad \} \quad -\infty, 2[$$

2

$$[2, +\infty[$$

$+\infty$

$$-2x + 7 - 2$$

$$-2x + 2 > 0$$

$$-2x > -2$$

$x \in \mathbb{I}$

$$2x - 7 - 2 > 0$$

$$2x - 6 > 0$$

$$2x > 6$$

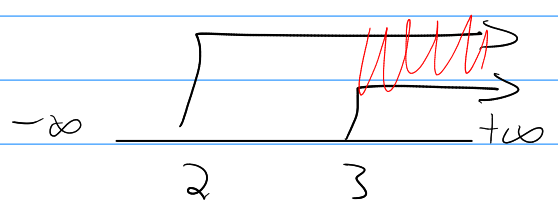
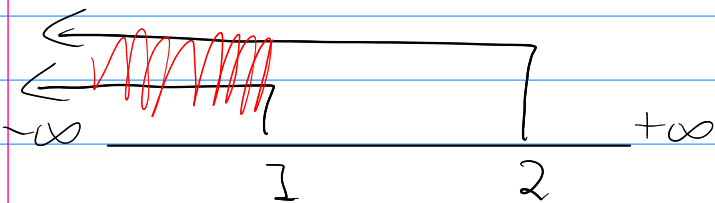
x > 3

$$]-\infty, 1[$$

$$\{3, +\infty\}$$

$$\} -\infty, 2[\quad \vee \quad \} -\infty, 1[$$

$$[2, +\infty[\cup]-\infty, -2]$$



$$S_I =]-\infty, I[$$

$$S_2 = \{ \} + \infty [$$

$$S_1 =]-\infty, 1[\quad \cap \quad S_2 =]3, +\infty[$$

$$S =]-\infty, 1[\cup]3, +\infty[$$

$\mathbb{R} \setminus F$ es positiva en $]-\infty, 1[\cup]3, +\infty[$

Se define la función de valor absoluto
 Como $f(x) = |x|$ con $f: \mathbb{R} \rightarrow [0, +\infty[$
 $f \geq 0$

Donde es negativa? $]1, 3[$

Intersecciones con los ejes?

$$x < 1 \quad x > 3$$

$$x = (-1, 0) \quad \cap \quad (3, 0)$$

$$Y: f(0) = \frac{|2 \cdot 0 - 7|}{2} = \frac{|-7|}{2} = \frac{7}{2}$$

$$Y: (0, 2)$$

$$f(x) = 2x + 1$$

Imagen ~~de $f(0)$~~ de 0

$$= 2 \cdot 0 + 1 = 1$$

Preimagen de 3

$$2x + 1 = 3$$

$$2x = 2$$

$$\boxed{x = 1}$$

f) $v: D_v \rightarrow \mathbb{R}$, tal que $v(x) = \frac{\sqrt{2|x| - 1}}{x^2 + x}$ 1 2

$$|x| \leq k, \text{ } k \text{ constante}$$

$$\leq \rightarrow -k \leq x \leq k$$

$$< \rightarrow -k < x < k$$

$$> \rightarrow -k > x > k$$

$$\geq \rightarrow -k \geq x \geq k$$

1 $\sqrt{2|x| - 1} \geq 0$

$$2|x| - 1 \geq 0$$

$$2|x| \geq 1$$

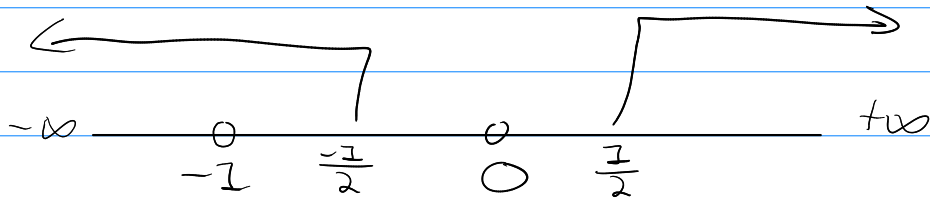
$$|x| \geq \frac{1}{2}$$

$$-\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$x \leq -\frac{1}{2} \quad \text{O} \quad x \geq \frac{1}{2}$$

$$\begin{aligned}
 2) \quad x^2 + x &\neq 0 & a \cdot b &= 0 \\
 x(x+1) &\neq 0 & a &= 0 \vee b = 0 \\
 x &\neq 0 & x+1 &\neq 0 \\
 x &\neq 0 & x &\neq -1
 \end{aligned}$$

$$\begin{array}{cccc}
 x \leq -\frac{1}{2} & , & x \geq \frac{1}{2} & , & x \neq 0 & , & x \neq -1 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
]-\infty, -\frac{1}{2}] & & [\frac{1}{2}, +\infty[& & x \neq 0 & & x \neq -1
 \end{array}$$



$$D_w \quad]-\infty, -\frac{1}{2}] \cup [\frac{1}{2}, +\infty[- \{-1\}$$

$$]-\infty, -1[\cup]-1, -\frac{1}{2}[\cup]\frac{1}{2}, +\infty[$$